ASCE Concrete Canoe — Structural Analysis

University of Virginia

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Abstract

This report documents the structural design of the optimised concrete canoe hull. Loads are derived from first principles, resolved into a governing bending moment and transverse turn moment, and checked against a composite transformed-section model. The lightest feasible design has a dry hull weight of **130.91 lb**. Every value herein is produced by the design program's live analysis engine; the section engine is independently validated (Sec. 13).

1 Design Objectives and Constraints

Quantity	Value	Source
Overall length	20.01 ft	optimised hull (Godzilla)
Maximum beam	20.69 in	optimised hull
Mold depth	15.07 in	sweep winner
Design draft	4.58 in	hydrostatic solve
Factor of safety	1.50	ACI 318-19 practice
Dynamic load factor	1.20	race dynamics
Freeboard target	6.0 in	team spec (ASCE 4-person)
Distinct mixes (cap)	2	batching constraint

2 Hull Geometry

The as-built section is taken directly from the optimised offset table (`off1.mlt`) at every station; no idealised constant-depth prism is used. Displaced volume is integrated from the real underwater sectional areas.

Length overall	20.01 ft
Maximum beam	20.69 in
Mold depth	15.07 in
Displaced volume at design draft	6.885 ft^3
Buoyancy at design draft	429.6 lb

3 Load Derivation

The hull self-weight per unit length $w(x)$ is built from the *real per-station transformed section area* times the local mix density (bottom-up over the pour zones), not a uniform prism. For each race

case the payload is applied as point loads at the paddler stations and balanced by a buoyancy line proportional to the local water-plane breadth. Shear and moment follow by integration:

$$V(x) = \int_0^x q(\xi)\, d\xi, \qquad M(x) = \int_0^x V(\xi)\, d\xi, \qquad q = q_{\text{buoy}} - w - w_{\text{pay}}. \tag{1}$$

The envelope is taken over all race cases (2-, 3- and 4-person) plus paddler-position variance. The governing bending case is **2F** at $x = 5.5$ ft, with $M_{\text{sag}} = 392$ ft-lb and peak shear $V_{\max} = 186$ lb.

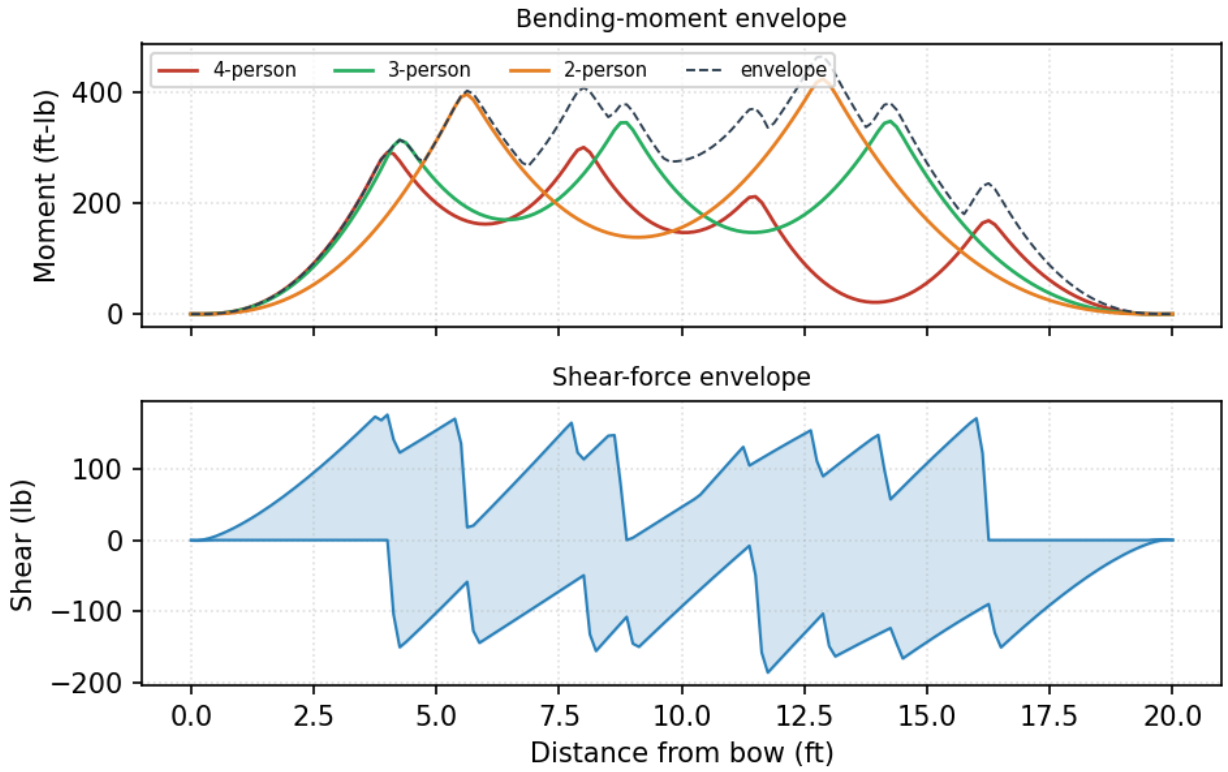


Figure 1: Bending-moment and shear-force envelopes across all race cases.

4 Governing Cross-Section (Worked)

At the governing station $x = 5.5$ ft the section is the three-zone I-beam. Using the transformed-section method with modular ratio $n_i = E_i/E_{\text{ref}}$ (reference $E_{\text{ref}} = 7.20 \times 10^5$ psi, the bottom-flange mix):

Zone	Bounds (in)	A_i (in^2)	E_i (psi)	n_i
Bottom flange (68.5 pcf)	0.00–4.00	6.319	7.20×10^5	1.000
Web (61.5 pcf)	4.00–10.00	4.508	2.99×10^5	0.415
Top flange (68.5 pcf)	10.00–15.07	3.802	7.20×10^5	1.000

Transformed neutral axis $\bar{y} = 5.965$ in above the keel; transformed second moment $I_{tr} = 299.2$ in^4 (in E_{ref} units). Bending stress at a fibre is $\sigma_i = n_i\, M\, (z - \bar{y})/I_{tr}$:

Each fibre is checked in *both* bending senses against the allowable for the sign of stress it actually carries. Under **sagging** ($M = 4705$ in-lb) the bottom fibre is in tension and the top

fibre in *compression*; under **hogging** ($M = 0$ in-lb) the senses swap. Checking a compression fibre against the tensile allowable would overstate its D/C by roughly the f'_c/f_t ratio (about 6× for these mixes) and is not a meaningful limit state.

Fibre	Sense	Stress	σ (psi)	Allowable (psi)	D/C
Bottom	sagging	tension	93.8	197.7	D/C=0.47 (gated)
Top	hogging	tension	0.0	197.7	D/C=0.00 (gated)
Top	sagging	compression	143.2	988.3	D/C=0.14 (advisory)
Bottom	hogging	compression	0.0	988.3	D/C=0.00 (advisory)

The two *tension* rows are the enforced limit states — the sweep gate rejects any design exceeding them. The compression rows are advisory: at this design the compression demand is far below f'_c (the D/C values above), so it is reported for visibility but does not affect feasibility.

The longhand above is pure bending. The gate's own utilisations at this station — which additionally include the neutral-axis shear via the principal-tension combination $\sigma/2 + \sqrt{(\sigma/2)^2 + \tau^2}$, and any mesh boost on the top fibre — are 0.527 (bottom) and 0.000 (top). The governing longitudinal value D/C = 0.527 carried into Sec. 6 and the compliance table is the larger of these two, so all three places report the same number against the same allowables.

5 Design Methodology

This design was checked by **LRFD**. The failure criterion is the Kupfer–Gerstle biaxial envelope of Sec. 8 under *either* methodology — LRFD and ASD change only how loads and capacities are factored, never what constitutes failure. The same per-fibre, per-station, per-load-case evaluation, the same per-zone allowables and the same degenerate-tip exclusion apply in both cases. **Load and Resistance Factor Design.** Factored load effects $\gamma_D D + \gamma_L L$ are compared against a design capacity ϕf_n with $\phi = 0.60$. The dead load is the hull self-weight including the stem-rocker cantilever tip mass; the live load is the paddlers, at the dynamic (impact-amplified) weights. The dynamic load factor 1.20 multiplies the crew weight *before* γ_L, so the effective factor on nominal crew weight is 1.20×1.60. **Resistance factor.** $\phi = 0.60$ is the ACI 318-19 Table 21.2.1(g) value for *plain* concrete (ACI 318-11 Sec. 9.3.5 used 0.55). It is deliberately *not* the 0.90 tension-controlled flexure factor: that value presumes a reinforced section in which steel carries the tension and the concrete's tensile strength is neglected outright. The governing check here is the concrete's own biaxial *tensile* capacity, which is the plain-concrete limit state — and the hull's mesh, not being conventional reinforcement, contributes nothing before the first cracking this check gates on.

Load case	Combinations (γ_D, γ_L)	Note
`2F`	(1.20, 1.60), (0.90, 1.60)	counteracting-dead alternative included
`2M`	(1.20, 1.60), (0.90, 1.60)	counteracting-dead alternative included
`3F`	(1.20, 1.60), (0.90, 1.60)	counteracting-dead alternative included
`3M`	(1.20, 1.60), (0.90, 1.60)	counteracting-dead alternative included
`4_coed`	(1.20, 1.60), (0.90, 1.60)	counteracting-dead alternative included

Counteracting dead load. Where the dead load *opposes* the live load, factoring D *up* understates the demand, and ASCE 7-22 Sec. 2.3.1 requires the dead factor to fall to 0.9 in that situation. That is the case here. At the governing station (102, load case `2F`) the unfactored components are $M_D = -26.5$ and $M_L = 281.0$ lb·ft — opposite in sign, the hull's self-weight hogging moment

relieving the crew's sagging moment. Applying $1.2D$ there would scale up a *relieving* action and leave the true demand understated, so the $0.9D$ alternative is enveloped alongside it and governs at that fibre. Both combinations are evaluated at every station and the worse of the two is taken; without the $0.9D$ case this design would have been checked against a demand lower than the code requires.

Stem overhang (tip cantilever). 1.05 in per end, *derived* from the hull geometry. This is the fore-aft run at each end where the maximum half-breadth falls below $1.5t$, so no shell section can exist and the material there is a slender cantilever tip. It is carried as un-buoyed dead weight beyond the supported span. The value is obtained by interpolating where the half-breadth crosses $1.5t$, so it is fixed by the hull rather than by the offset file's station count — the same hull returns the same overhang at 41, 81, 161 and 321 stations. It is *not* a design variable: because the tip mass raises hogging at the ends, and end hogging opposes the crew's midship sagging, a longer overhang *lowers* the governing D/C. Treating it as a free parameter would let the optimiser buy structural margin with tip mass, which is a far worse weight-per-margin trade than mix density or wall thickness. It is therefore measured, never tuned. Note this is not the same geometry as the CAD stem sweep, which raises existing material upward *within* the hull over the same length; the structural model instead extends the beam and places the mass *outside* it. The two share only the length, and the structural treatment is the bounding one. The beam model appends the overhang as whole stations, so its effect is a step function of the analysis grid rather than a continuous one: an overhang shorter than about one and a half station spacings contributes no mass at all. At the resolution used here that is the case, so the tip mass ($\approx$0.1 lb) is neglected. That is conservative — tip mass relieves the midship sagging that governs — but it is a consequence of the discretisation, not a modelling choice, and no smooth relationship between overhang and utilisation should be read into these results. The acceptance margin of Sec. 6 ($\mathrm{D/C} \leq 0.95$) is applied identically under both methodologies: it corrects for a numerical non-convergence in the utilisation, which is unrelated to how loads and resistances are factored.

6 Structural Checks

The transverse-hoop, longitudinal-bending, shear and punching checks below are *enforced* by the optimiser — a design failing any of them is infeasible and cannot win. The closing torsion/lateral figures are *advisory* conservative bounds, reported for visibility but not gated (see that subsection).

Acceptance margin ($\mathbf{D/C \leq 0.95}$). The winning design is required to stay below 0.95 on the governing biaxial check, not below 1.0. This is a margin against a *numerical* limitation, not additional structural conservatism — the factor of safety covers the physics and is already inside every allowable quoted here. The hull weight converges quickly with longitudinal station count (0.58% between 161 and 321 stations), but the Kupfer utilisation that decides feasibility does not: measured across five designs on this hull it drifts 1–2% for every doubling of the station count, with no plateau observed out to 641 stations. A design accepted at 1.0 on the 161-station analysis grid is therefore not demonstrably feasible — one refinement step can reject it, and on this hull did (0.9957 at 161 stations, 1.0040 at 321). Accepting at 0.95 buys roughly 2.4 doublings of headroom at the worst drift rate observed, so the verdict holds to about 830 stations; the accepted design is additionally re-checked at 321 stations and confirmed below 1.0 there.

Reading the D/C ratios. Every utilisation ("D/C") quoted here and in the figures is demand divided by the *allowable*, and the allowable already carries the factor of safety: allowable = ultimate strength/FoS with FoS = 1.50. A D/C of 1.0 is therefore the *allowable* limit, not incipient failure — a value of, say, 0.90 already sits below a capacity that has been reduced by the full 1.50×

safety factor, so the true margin to ultimate is larger still.

6.1 Transverse (hoop) bending — transformed section

In a turn of radius R at speed V the whole hull is accelerated laterally, so the centripetal load per unit length follows the local mass and bends the section about its vertical centre-plane. Because that load follows the *same* mass distribution as gravity and is reacted by the *same* waterplane, the balanced lateral turn moment is the vertical bending envelope scaled by the turn load factor:

$$M_{\text{lat}}(x) = k_{\text{lat}}\,|M_{\text{vert}}(x)|, \qquad k_{\text{lat}} = \frac{V^2}{gR}. \tag{2}$$

This is a genuine per-station *distribution*: it vanishes at the bow and stern (where $M_{\text{vert}} \to 0$), not a single scalar applied at every station.

Reaction approximation. The lateral load is reacted here on the waterplane breadth (as buoyancy is), whereas the true hydrodynamic side force acts on the submerged lateral profile. For this hull the two distributions differ in shape, but the resulting peak M_{lat} agrees to within 1%, so the idealisation does not move the check.

The section is vertically zoned, so the check uses a *transformed* lateral section: each pour zone's outboard fibre is checked against *that zone's own* tensile strength, and the governing fibre is the soft light web at the hull's widest point. Governing at $x = 12.9$ ft in the **61.5 pcf web** (zone 2), D/C = 0.100 — well inside unity. The hull is far stiffer laterally than vertically (its lateral second moment is several times the vertical), and the turn load is only $k_{\text{lat}} \approx V^2/gR$ of gravity, so longitudinal bending — not the turn — governs. Fibre-by-fibre (homogeneous reference D/C, for comparison): Bottom flange 0.04, Web 0.18, Top flange 0.04.

6.2 Longitudinal bending

Peak D/C = 0.527 at $x = 5.5$ ft (worked in Sec. 4). Governing fibre: the **bottom fibre in tension under sagging**, at $\sigma = 93.8$ psi against the tensile allowable $f_t = 197.7$ psi (bending longhand; the gate value above also carries the neutral-axis shear term). Tension-controlled — compression demand at both fibres is well inside f'_c and does not govern.

6.3 Shear

Peak shear $V_{\max} = 186$ lb; shear stress is well below the tension check and does not govern.

6.4 Torsion and lateral — advisory (not gated)

A one-sided paddle stroke twists the hull. The internal torque is an equilibrium distribution $T(x)$ (the stroke torque reacted by the waterplane, so it closes to zero at the free ends), and the shear follows open-section St. Venant theory, $\tau = |T(x)|\,t/J$ with $J = t^2A/3$. The peak is $\tau = 345$ psi at $x = 6.0$ ft against an allowable 98 psi, D/C = 3.53. **This bound is deliberately conservative and is NOT enforced.** The open-section J ignores the warping restraint of the closed stems and the gunwale flanges (and the water's support), so it overstates the true torsional shear by roughly an order of magnitude — which is why hulls of this geometry are built and paddled without torsional distress. A physically meaningful torsion gate would need a warping-inclusive model; until then this figure is reported, not applied. The companion lateral-bending advisory (the same distributed $M_{\text{lat}}(x)$ treatment as the hoop check above) is D/C = 1.63.

6.5 Paddler load split (knees vs toes)

A kneeling paddler carries weight on the knees and the toes. This model uses **two different splits**, one per limit state, because the conservative direction is not the same for both:

Limit state	Knees	Toes	Basis
Bending / shear envelopes	100%	0%	team-measured
Punching (local two-way)	100%	0%	conservative bound

Measured data, not a literature constant. The 100/0 bending split is a *team measurement*: a paddler was weighed in racing kneeling position with the knees and the toes bearing on separate scales, and the knee reading taken as a fraction of body weight. **Sample size** $n = 1$ — this is a single measurement of one paddler, not an established biomechanical constant, and it is reported here with the same caveat as the small-n mix regression (Sec. 9). It should not be cited as a general figure. The split is a configuration field (`knee_frac_bending` in `canoe_suite.cfg`), so measuring more paddlers updates every downstream number without a code change. **Why punching does not use it.** Punching is a *local* contact check: crediting 0% of the weight to the toes would reduce the load on the knee patch and make the check less severe on the strength of a single measurement. The punching basis therefore keeps 100% on the knees — no toe relief — so the two-circle result in the next subsection stands on the conservative bound regardless of how the measured split later moves. **Currently inactive for bending.** `ADVANCED_PADDLER_LOADING` is *off* for this run, so the envelopes in this report apply each paddler's *full* weight as a single point load at the knee position — no toe split. Both models apply the *same* total paddler weight, verified to machine precision on every grid resolution; they differ only in how that weight is distributed, the point-load model simply concentrating it at the knee. **Neither model is uniformly conservative.** Moving weight 17 in aft to the toes lowers the *sagging* moment and the peak shear, but *raises* the hogging moment, because the toe loads sit further outboard of the supports. Which effect wins depends on the seating and the hull, so the two models cannot be ranked once and for all — and the governing D/C can move either way. The point-load default is retained because it is the simpler, more directly checkable idealisation, not because it is a bound. Any design decision that turns on the difference must re-run *both* models on the design in question rather than assuming a direction.

6.6 Punching shear (kneeling paddler)

Step 1 — rigid 3 in square patch (conservative bound). A paddler kneels on a single 3 in square patch on the hull *bottom* (ACI 318-19 §22.6.5.2). The concentrated weight is taken conservatively as the full heaviest paddler, $P = 175$ lb, on one patch. Demand $v_u = P/(b_0 t)$ with critical perimeter $b_0 = 4(c + d) = 13.00$ in (effective depth $d = 0.250$ in — the reinforcing mesh on its buildable layer interface, set by the construction schedule rather than assumed at $t/2$; see the mesh-depth sensitivity below):

$$v_u = \frac{175}{(13.00)(0.375)} = 35.9 \text{ psi}. \tag{3}$$

$f'_c = 1482$ psi is the **raw** compressive strength of the **bottom-flange (68.5 pcf) mix** — the concrete at the kneeling surface, not the top mix. It is the *unfactored* strength on purpose: the ACI capacity equations below are written in terms of f'_c and the factor of safety is applied once, afterwards, to their result. ACI gives three concrete capacities; the *smallest governs* (which one

depends on geometry, so any single formula is non-conservative):

$$\text{(a) } v_c = 4\lambda\sqrt{f'_c} = 115.5 \text{ psi} \tag{4}$$

$$\text{(b) } v_c = \left(2 + \tfrac{4}{\beta}\right)\lambda\sqrt{f'_c} = 173.3 \text{ psi} \quad (\beta = 1,\ \text{square patch}) \tag{5}$$

$$\text{(c) } v_c = \left(2 + \tfrac{\alpha_s d}{b_0}\right)\lambda\sqrt{f'_c} = 80.0 \text{ psi} \quad (\alpha_s = 40,\ \text{interior}) \tag{6}$$

Equation **(c)** governs at 80.0 psi. For this thin shell the effective depth d is small relative to b_0, so the $\alpha_s d/b_0$ term is small and (c) falls below (a). The paddler kneels on the flat hull bottom, well inside the panel, so the interior $\alpha_s = 40$ applies (an edge condition, $\alpha_s = 30$, would be lower still). Allowable $= v_c/\text{FoS} = 53.3$ psi, giving D/C $= 0.67$ — **PASS**.

Mesh depth follows the layer schedule, not a $t/2$ assumption. The effective depth above is $d = 0.250$ in: the reinforcing mesh embedded on a real interface of the 3-pass build. Because d enters the governing $\alpha_s d/b_0$ capacity term, *which* inter-pass interface the mesh lands on changes the punching D/C. The single-square bound at the two candidate interfaces and at the legacy mid-shell assumption:

Mesh placement	d (in)	D/C	
After pass 1 (shallow, $t/3$)	0.125	D/C = 0.81	PASS
Mid-shell (legacy $t/2$)	0.188	D/C = 0.73	PASS
After pass 2 (deep, $2t/3$) — design	0.250	D/C = 0.67	PASS

Placing the mesh a pass too shallow lifts D/C from D/C = 0.67 to D/C = 0.81 ($\approx$ 20% less margin). A crew member embedding mesh after the *first* pass is the intuitive move, so the deep interface is written as a **binding** construction requirement (Sec. 15), not a recommendation.

Step 2 — two 3 in-diameter circular knee contacts. Paddlers are lifted and *placed* onto both knees; nobody steps in heel-first, so the single-heel case that governs some designs does not arise here. That placement procedure is a binding compliance note (Sec. 15), because it is what licenses this model. The weight is split 50/50 across the two knees, 175 $\rightarrow$ 87.5 lb per knee. No credit is taken for the 20% toe share of the 80/20 biomechanics split — the full body weight is kept on the knees, so the split is conservative on load. The contact is circular, so the ACI critical perimeter at the $d/2$ offset is $b_0 = \pi(d_{\text{load}} + d)$, *not* the square $4(c + d)$: 10.21 in versus 13.00 in, i.e. 21.5% *less* perimeter for the same contact width. This works *against* the design — it raises stress per unit load and partly cancels the load split. Isolating the two effects: the same 3 in circle carrying the *full* 175 lb gives D/C = 0.80 (worse than the square's D/C = 0.67); halving the load then brings it to D/C = 0.40. Net factor = 0.50/0.785 = 0.637. **Perimeter interaction.** Two nearby contacts cannot be treated as independent half-load punches if their critical perimeters merge, since a merged section can be worse per unit load. Each critical circle has radius $(d_{\text{load}} + d)/2 = 1.625$ in, so the perimeters merge only if the knee spacing drops below 3.25 in. At the assumed centre-to-centre spacing of 8.0 in (hips-width kneel) the two perimeters are well clear of each other, so they act as two independent single-knee punches and the per-knee check above applies. The check re-evaluates this automatically if the spacing is changed. Governing capacity equation **(c)** at 86.0 psi, allowable 57.4 psi, demand 22.9 psi: D/C = 0.40 — **PASS**.

Step 3 — legal knee pad on top of the two-circle model. ASCE rules cap knee-pad thickness at 0.5 in. A pad spreads each contact to an effective diameter $d_{\text{load}} = 3 + 2t_{\text{pad}}$. With the full legal 0.500 in pad, $d_{\text{load}} = 4.00$ in and D/C = 0.33; with a realistic thinner 0.250 in pad,

$d_{\text{load}} = 3.50$ in and D/C = 0.36. The pad's benefit is real but modest: growing b_0 also shrinks the $\alpha_s d/b_0$ term in equation (c), so the allowable falls slightly (57.4→52.9 psi) as the demand falls. The pad is therefore reserve, not the basis.

Progression and adopted basis.

Step	Contact model	b_0 / load	D/C
1	rigid 3 in square, full load	13.00 in / 175 lb	D/C=0.67 PASS
2	two 3 in circles, split, no pad	10.21 in / 88 lb	D/C=0.40 PASS
3a	+ 0.250 in pad	11.78 in / 88 lb	D/C=0.36 PASS
3b	+ 0.5 in legal-max pad	13.35 in / 88 lb	D/C=0.33 PASS

The documented design basis is the rigid single-square bound (Step 1), at D/C = 0.67. The crudest model already clears, so the design basis takes *no* credit for the kneeling refinement at all — Steps 2 and 3 are reported as reserve, and none of the contact, spacing or pad assumptions are load-bearing.

7 Multi-Zone I-Beam Pour

The design pours a three-zone vertical I-beam from **2 distinct mixes**: 68.5/61.5/68.5 pcf (bottom flange / web / top flange). The dense flanges sit at the extreme fibres and the light web spans the neutral axis; the same stack is used along the whole hull length (no longitudinal zoning).

Zone	Density (pcf)	f'_c raw (psi)	f_t raw (psi)	f'_c/FoS allowable (psi)	f_t/FoS allowable (psi)
Bottom flange	68.5	1482	296.5	988	197.7
Web (core)	61.5	353	70.7	236	47.1
Top flange	68.5	1482	296.5	988	197.7

Raw strengths feed code equations written in terms of f'_c (the ACI $\sqrt{f'_c}$ punching terms and the ACI modulus), which apply the factor of safety themselves. The allowable columns (FoS = 1.50) are what member D/C ratios divide by. Using an allowable where a raw strength belongs applies the factor twice.

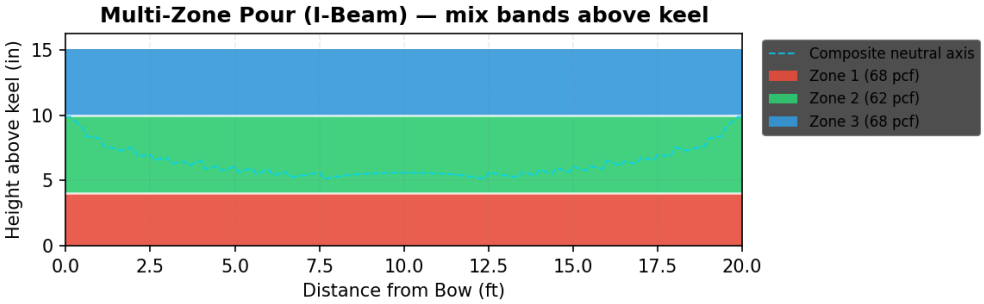


Figure 2: Pour-zone section: three density bands with the neutral axis overlaid.

7.1 Construction Layer Schedule (through-thickness)

The pour zones above are *vertical* bands that set *which* mix sits at a given height. The **construction layers** are the *orthogonal* through-thickness dimension — *how many* troweled passes build the wall up to its thickness. Every point of the shell is applied in **3 successive equal passes** of 0.125 in

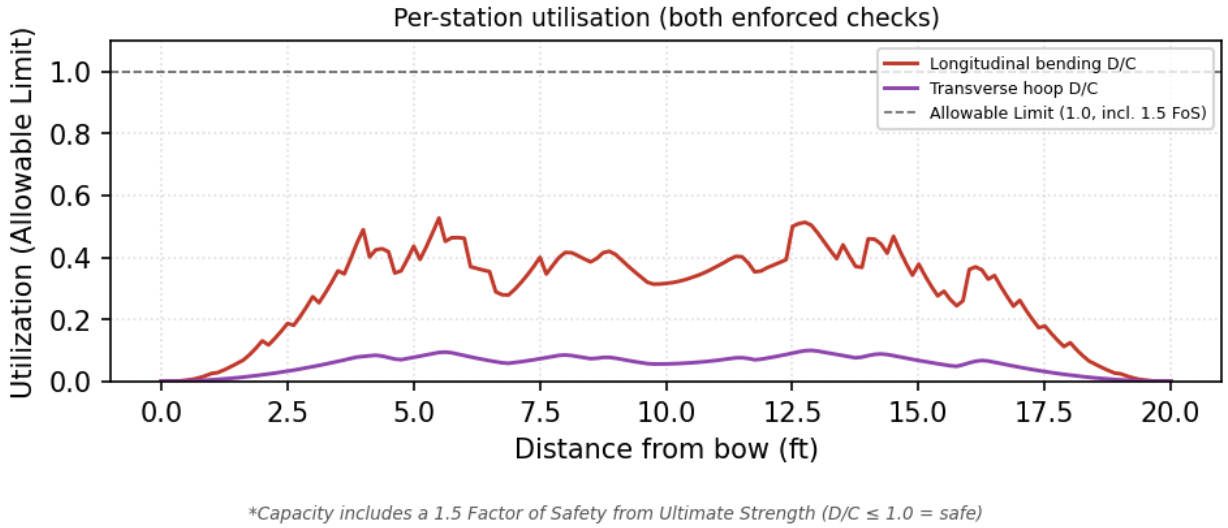


Figure 3: Per-station demand/capacity for both enforced checks.

(= 4/32 in) (total 0.375 in), the same count everywhere so the thickness is uniform. Each pass carries the mix of whatever pour band it crosses (mix per height from the zone table above); the layer count itself does not change from zone to zone.

Each 0.125 in pass clears the 0.0625 in (1/16 in) hand-trowel minimum, so the schedule is buildable as stated.

Uniform-thickness check. The structural weight model is a thin-wall shell (perimeter $\times t$); the CAD export solidifies any near-keel band that would be thinner than the shell. On this hull the as-built ring weighs 131.7 lb against the modelled 133.7 lb (-1.5%; 0 of 39 stations solidify $> 5\%$): the two agree and the built shell is not heavier than analysed, so the uniform 0.375 in premise holds.

8 Combined Biaxial Stress — Kupfer-Gerstle Envelope

The governing tensile check is a biaxial concrete failure criterion, not the maximum-principal-stress (Rankine) rule. Rankine ignores that concrete's tensile capacity *degrades under transverse compression*, and this hull sits squarely in the tension–compression quadrant: longitudinal bending puts a fibre in tension while the centripetal (hoop) moment $M_{\text{lat}}(x) = k_{\text{lat}}\,|M_{\text{vert}}(x)|$ compresses that same fibre transversely. There Rankine is unconservative exactly where the design governs.

At every fibre z of every station, under each individual load case (2/3/4-person and their position-variance perturbations — not the envelope), the coexisting state is σ_x (longitudinal bending $n\,M_{\text{vert}}(z-\bar{y})/I_{tr}$), σ_y (transverse hoop $n\,M_{\text{lat}}\,y/I_{lat}$), and $\tau_{xy} = V\,Q(z)/(I_{tr}\,2t)$. Principal stresses $\sigma_{1,2} = \frac{\sigma_x+\sigma_y}{2} \pm \sqrt{(\frac{\sigma_x-\sigma_y}{2})^2 + \tau_{xy}^2}$ feed the Kupfer criterion, quadrant-aware (each pour zone against its *own* f_t, f_c'):

$$\text{T–T } (\sigma_2 \geq 0): \quad \sigma_1/f_t \leq 1 \tag{7}$$

$$\text{T–C } (\sigma_1 > 0 > \sigma_2): \quad \sigma_1/f_t + 0.8\,|\sigma_2|/f_c' \leq 1 \tag{8}$$

$$\text{C–C}: \quad \text{biaxially strengthened to } \sim 1.16 f_c' \text{ (crushing checked separately vs. uniaxial } f_c') \tag{9}$$

The tension–compression line $\sigma_1/f_t = 1 - 0.8\,|\sigma_2|/f'_c$ is the Kupfer, Hilsdorf & Rüsch (1969) experimental fit as adopted by CEB-FIP Model Code 1990 (the 0.8 slope is theirs, not invented here). **Torsion is excluded** from τ_{xy}: this suite's torsion figure is an open-section St-Venant bound that overstates by $\sim 11\times$ (it ignores warping restraint), so folding it into the fibre state would be nonsense.

How much the biaxial interaction adds, for the winning design (design D/C, FoS already inside):

Check	D/C
Independent longitudinal (bending)	D/C = 0.527
Independent transverse (hoop)	D/C = 0.100
Rankine, combined state (max principal)	D/C = 0.894
Kupfer biaxial (governing gate)	D/C = 0.903

Governing fibre: station index 102, $z = 15.07$ in, race case index 1, **tension-compression** quadrant ($\sigma_x = 159.0$, $\sigma_y = -9.8$, $\tau = 0.5$ psi).

Honest limits. Concrete Damaged Plasticity and Willam-Warnke reproduce the biaxial (and triaxial) surface more accurately, but they are nonlinear-FEA constitutive models — the right tool for a future ANSYS Mechanical study, not a closed-form gate inside a parametric sweep. Kupfer is the accuracy ceiling for the closed form. Critically, f_t and f'_c here come from an $n = 2$ density–strength regression, so the criterion's sophistication exceeds the precision of its inputs: a better criterion on uncertain data is still the right call, but the combined D/C should not be read as more accurate than the two lab breaks behind it allow.

9 Mix Regression

Allowable stresses come from the team's density–strength regression (source: `mixes`, $n = 3$ lab points, tag `v12-2026-07-11`):

$$f'_c = 161.29\,\rho - 9566.0, \qquad f_t = 32.26\,\rho - 1913.2 \text{ (psi, } \rho \text{ in pcf)} \tag{10}$$

Operating densities [61.5, 68.5] lie within the lab data range [60.0, 80.0] pcf (interpolation). The regression predicts zero tensile strength below $\rho = 59.3$ pcf (the tensile x-intercept), which sets the practical light-mix floor.

9.1 Mix Provenance: Batched vs. Predicted

The optimiser searches a continuous density spectrum, so a winning density need not correspond to a mix that has ever been batched. Each zone below is therefore marked either **on record** (a mix in the Mix Lab) or **predicted** (a density the regression extrapolated to). For predicted densities the nearest real mixes on either side are given, together with the *reach*: how far the prediction sits from the closest data point. A short reach between two bracketing mixes is an interpolation; a long reach, or no bracket at all, is an extrapolation the mix team must verify before this design can be relied on.

Zone	ρ (pcf)	f'_c (psi)	f_t (psi)	Provenance
bottom flange	68.5	988	198	predicted; between 60.0/70.0, reach 1.5
web	61.5	236	47	predicted; between 60.0/70.0, reach 1.5
top flange	68.5	988	198	predicted; between 60.0/70.0, reach 1.5

Mixes currently on record: 50.9, 51.2, 55.5, 56.2, 60.0, 70.0, 80.0 pcf. Quoted f'_c and f_t are allowables, with the factor of safety already applied. This design targets mixes that do not yet exist; the predicted densities must be batched and tested before the quoted capacities can be claimed.

10 Transverse Stability

Transverse metacentric height is $GM_T = KB + BM_T - KG$. The first two terms are hull FORM; only KG depends on where the concrete actually sits. A GM_T quoted without stating its KG basis is therefore uninterpretable — the same hull yields materially different numbers under the three conventions below, and they must not be compared across bases. For this hull $KB = 5.72$ in and $BM_T = 8.78$ in, giving a form ceiling of 14.50 in against which every KG is charged. The poured shell's own centre of gravity is $KG_{hull} = 6.28$ in.

Convention	KG basis	GM_T
Form-only	waterline (shape alone)	9.92 in
Bare hull	real zoned-pour shell CG	8.22 in
Loaded (racing)	shell + crew, mass-weighted	-2.99 in

The form-only figure is the one blind to mix distribution: an I-beam with heavy top flanges scores identically to a uniform light hull. It is reported because it is the convention a published hull-form comparison is most likely to use, NOT because it describes this canoe's behaviour. The loaded value is the only one that describes a capsize.

Crew centre of gravity. The 620 lb crew outweighs the 130.9 lb hull by roughly 4.7 to one, so the crew's centre of gravity, not the concrete, sets the combined $KG = 17.49$ in and with it the entire loaded stability result. It is computed from segmented anthropometry rather than assumed. A kneeling paddler is modelled as two groups, because only one of them rotates:

$$KG_{paddler} = f_{legs}\, h_{legs} + f_{HAT}\,(h_{hip} + L_{HAT} \sin\theta)$$

where f are segment mass fractions, h_{hip} is the hip-joint height above the hull floor, L_{HAT} the hip-to-CG distance of the head-arms-trunk group, and θ the torso angle from horizontal. Only the torso, head and arms swing about the hip when a kneeling paddler leans; the shins are flat and the thighs do not rotate. Crew KG is the mass-weighted mean over the paddlers, which is why the coed seating ARRANGEMENT cannot affect it — a property of the sum, not an unhandled case. Segment masses are sex-specific from Plagenhoef, Evans and Abdelnour (1983); segment lengths follow Drillis and Contini (1966) as tabulated in Winter. L_{HAT} is not asserted but CALIBRATED, by solving for the offset that reproduces the published standing whole-body CG ($0.565H$ male, $0.545H$ female), giving $0.159H$ and $0.148H$. As an independent check, the same length data predicts a kneeling-height/stature ratio of 0.715 against ANSUR II's measured 0.709 — agreement to 1%.

Basis for this run: biomechanical model, 4_coed, posture B_upright_kneel, torso 70 deg, hip from posture preset (NOT measured), giving $KG_{crew} = 19.86$ in above the keel.

This REPLACES the flat 12 in assumption used previously, and it is not a small correction: loaded GM_T moves from 3.49 in to -2.99 in, a change of -6.49 in. Every loaded stability figure this suite reported before this model was optimistic.

POSTURE DOMINATES. Two kneels are plausible and they are not close: hips resting back on the heels puts the hip joint near $0.10H$, while an upright kneel with the thighs vertical puts it near $0.245H$ — about eight inches apart. Sweeping the torso angle across its entire plausible

range moves the crew CG by roughly two inches. The posture, not the lean, decides the answer:

Posture	θ	KG_{crew}	loaded GM_T
A heels sitting	45°	11.01 in	4.31 in
A heels sitting	90°	13.00 in	2.67 in
B upright kneel	45°	18.28 in	-1.69 in
B upright kneel	90°	20.27 in	-3.33 in

The upright kneel is the default because the team's own measurement supports it: at a measured 90/10 knee/toe load split with the toes 17 in aft of the knee, horizontal equilibrium about the knee contact places the combined CG within about 1.3 in of the knee. Heels-sitting would put roughly two thirds of the load on the toes, which is not what was weighed. That measurement is $n = 1$.

Loaded GM_T is NEGATIVE. Over the default posture the metacentre sits BELOW the combined centre of gravity: the centre of buoyancy is not above the centre of mass in the racing condition, and the hull is not self-righting with the crew aboard. This is reported rather than tuned away, and it is consistent with the 2025 UVA report, which stated the same result for last season's hull and accepted it as the best attainable under the freeboard and width constraints. The operational consequence is plain: THE CANOE IS STABLE BECAUSE THE PADDLERS ACTIVELY BALANCE IT, not because the hull rights itself. Kneeling places roughly 620 lb of crew high above a shallow hull, which is the dominant term and is inherent to the boat class rather than a defect of this design.
This also settles the interpretation of the peer figures below. A published GM_T near 10 in cannot be a LOADED value for a canoe of this class — it would require a combined KG no kneeling crew can produce — which supports reading them on a hull-form or bare-hull basis.

ACTION, not caveat. This is an anthropometric ESTIMATE. Because hip height dominates and hip height is a single tape-measure reading on a kneeling paddler, MEASURING IT WOULD SUPERSEDE THIS ENTIRE MODEL. Until then the posture preset stands in for the measurement, and a direct determination of a kneeling paddler's CG would retire the estimate altogether. The assumption is improved here, not eliminated.

Peer comparison. Florida's 2026 report publishes $GM_T = 10.4$ in (Reptilia) and $GM_T = 10.9$ in (Tempest). That table lists GM_T beside waterline length, length-to-beam ratio, entrance angle, transom width and drag — all hull-FORM properties — and compares the value across two hull SHAPES, which is only meaningful on a form basis. We therefore read their figure as form-only or bare-hull and compare it to our form-only 9.92 in, a gap of 4.6–9.0%. Their loading basis is not stated in the source, so this comparison is a reasoned reading and not a verified apples-to-apples match; it must not be compared against our loaded value.

Because the gap is on a form basis, it is a HULL-SHAPE gap and cannot be closed by redistributing concrete. $BM_T = I_{wp}/V = 8.78$ in is the ceiling, set entirely by waterplane distribution: our hull was generated by optimising RESISTANCE, with stability never entering the objective, whereas a flat-bottomed form raises I_{wp} deliberately. Whether stability should become an explicit constraint in future hull generation is a next-season question, recorded here and not acted on in this analysis.

Structural tension — the top flange. The I-beam section places DENSE concrete in the top flange because that is where bending capacity is earned; that is precisely the placement that raises KG and costs stability. The two objectives pull on the same variable, so the trade is measured on this winner: dropping the top zone from 68.5 to 61.5 pcf moves loaded GM_T by -0.02 in and weight by -3.70 lb, at a Kupfer utilisation of D/C=3.036 (INFEASIBLE — the capacity is not available

to trade).
That GM_T movement of 0.02 in is SMALLER than the 7.64 in spread the posture band alone spans. Redistributing concrete is therefore not a meaningful stability lever on this hull: the crew dominates KG, so the top flange should be sized by bending capacity, and stability sought from hull FORM (BM_T) or from a measured crew CG instead.

11 Strength Basis and Winning Load State

Design strengths come from the density–strength regression currently IN FORCE, refitted from the Mix Lab table on load and after every lab edit. Source: `mixes`, n=3; no 3-point-bending (MOR) points exist, so the bending gate falls back to split-tensile.
The tensile tier in force is **split-tensile**, and the density search floor derives from it: **59.3 pcf** (split-tensile (n=3): zero-strength at 59.31 pcf, 1 psi at 59.3 pcf). It is NOT taken from the compressive fit — the two cross zero at different densities, and every bending gate here limits on TENSION, so a density with compressive strength but no tensile capacity cannot pass whatever its f'_c.
STATED MODELLING CHOICE — MOR in a biaxial criterion. Where a flexural (MOR) fit exists it is preferred for bending, because the gate computes $\sigma = Mc/I$, the extreme-fibre stress in flexure, which is precisely what a 3-point bending test measures; split-tensile is back-calculated from a diametral load and the compressive route is a correlation. MOR is therefore the least-converted measurement for a bending limit state. It is also the most PERMISSIVE tier — for the same mix it typically exceeds split-tensile, so preferring it RAISES allowables and yields a lighter design. Further, the Kupfer criterion uses f_t as the tensile capacity in a COMBINED stress state, whereas MOR is calibrated to bending alone; using it there is reasonable but slightly permissive. Both are deliberate, and are recorded here for the same reason as the M_{lat} reaction approximation and the open-section torsion bound.

Winning seat positions [ft from the bow]. Recorded so the winning load state can be REPRODUCED exactly — a gate-table probe that guesses the seating changes the moment envelope enough to make its answer meaningless.

Race	Seats [ft]
midrule 2M	5.594, 12.872
2F	5.630, 12.841
3M	4.250, 8.823, 14.238
3F	4.251, 8.827, 14.232
4_coed	4.020, 7.990, 11.499, 16.236

12 Wave Resistance

Wave resistance is a hydrodynamic quantity outside the structural scope of this analysis. A resistance figure is reported here only when validated hydrodynamic (CFD) data is supplied for the hull; in its absence none is printed, because an unvalidated number would misrepresent the design. This is a deliberate reporting choice: the report states only what the analysis has established.

13 Section-Engine Validation

Every stress in this report is produced by one section engine: it integrates the hull offsets into area, neutral-axis height and second moment, and for a multi-material pour transforms each zone by its

modular ratio $n_i = E_i/E_{\text{ref}}$ before locating the composite neutral axis. The engine is verified below against three closed-form results, each reproducible by hand and using no external or measured data.

13.1 Base integration against exact theory

A thin wall of height H has second moment $I = \frac{1}{12}tH^3$ about its mid-height; the hull section is two such walls (port and starboard), so $I = 2 \cdot \frac{1}{12}tH^3$. For $H = 8.122\,\text{in}$ and $t = 0.75\,\text{in}$ the engine returns $I = 66.97\ \text{in}^4$ against the exact 66.97 — **0.000% error**. This confirms the raw area and second-moment integration of the real offsets.

13.2 Transformed neutral axis against its closed form

Splitting the same wall into two equal-height zones with modular ratio $n = 2$ shifts the neutral axis toward the stiffer material. Because the area per unit height is constant, the exact transformed NA is

$$\bar{y} = \frac{c^2/2 + n(H^2 - c^2)/2}{c + n(H - c)}, \qquad c = \tfrac{H}{2}. \tag{11}$$

The engine returns $\bar{y} = 4.7378$ in against the closed form 4.7378 in — **0.0001% error**. This verifies the modular-ratio bookkeeping the multi-zone I-beam depends on: the neutral axis it locates is the true transformed centroid, not an approximation.

13.3 Homogeneous limit

When every zone shares one modulus the transform must collapse exactly to the plain Mc/I section. Setting both zone moduli equal, the engine reproduces the base second moment to 9.55×10^{-13}% — a necessary consistency condition, satisfied to machine precision.

Three analytic self-checks — exact theory for the base integration, a closed-form transformed neutral axis, and the homogeneous limit — verify the section engine from first principles. Every stress in this report flows through it.

14 Search Space and Boundary Check

A winning design is only meaningful together with the space it was selected from: refining the increments inside a range that is itself too narrow cannot find an optimum that lies outside it. The sweep therefore records the ranges and increments actually searched:

Variable	Range	Increment	Count
Wall thickness	0.375–0.375 in	buildable (1/32″; 0.5 mm)	1
Mold depth	12.57–17.07 in	0.500 in	9
Mix density	60.0–70.0 pcf	0.50 pcf	153 combos
Longitudinal	161 stations	(accuracy; native 41)	interpolated above native

Winner on a search boundary. The optimum sits on an edge of the searched space for:

- **thickness** at its min (0.375) — a *physical* limit (min buildable trowel); widening the range would not move it.

15 Safety Factors and Compliance

Every D/C entry below is demand over the *allowable* (ultimate strength / 1.50 FoS); a value ≤ 1.0 is compliant *with the safety factor already applied.*

Check
Dry hull weight
extbfKupfer-Gerstle biaxial (GOVERNING)
legacy longitudinal (bottom fibre in tension under sagging), not
Transverse hoop
Punching shear
Factor of safety
Loaded freeboard (4-person)
Transverse stability GM_T (form-only)
— loaded, real KG (§10)

Seating arrangement. Every physically possible seat ordering was enveloped for every race (2F: 2, 2M: 2, 3F: 6, 3M: 6, 4_coed: 24), so the design does not depend on who sits where.

BINDING — punching-shear design basis. The punching D/C above is carried by the shell itself under the conservative single-square bound, so the kneeling refinements impose no operational requirements. The mesh-placement and mix notes below remain binding — both set the numbers the check was run at:

- **Reinforcement mesh depth.** The mesh is placed at the 0.250 in interface — after pass 2 of 3, the deeper of the two inter-pass interfaces. Embedding it one pass shallower (after pass 1, at 0.125 in) drops the effective depth d below the design 0.250 in and *overstates* punching capacity (D/C would rise to D/C = 0.81). Embedding mesh after the first pass is the intuitive move, so this is an explicit build instruction, not a footnote.
- **Bottom-flange mix strength.** $f'_c \geq 1482$ psi at the kneeling surface (the 68.5 pcf tension mix). The check is against the *bottom* mix; substituting a lighter mix in the bottom flange invalidates this row.
- **Knee pads.** Not required by the check — the basis takes no pad credit. Pads worn for comfort are unquantified reserve (they would take D/C to D/C = 0.33), up to the ASCE 0.5 in legal cap.

Loaded freeboard — derivation. ASCE measures freeboard *under four-person load*, so the compliance figure is not the geometric mold-depth-minus-design-draft value. The loaded waterline is root-found on the real hull offsets: for a trial waterline z each station's submerged area is integrated over depth and those areas are integrated along the hull,

$$\nabla(z) = \int_0^L \left[\int_0^z 2\, y(x,\zeta)\, d\zeta \right] dx, \qquad \text{solve } \rho\, \nabla(z_{wl}) = W_{\text{loaded}}. \tag{12}$$

The offsets $y(x,z)$ are the *same* per-station table the section engine uses (no separate hydrostatic hull model). Loaded weight is the hull at 130.91 lb plus the four-person crew (175+165+145+135 lb) = **750.91** lb, with $\rho = 62.4$ lb/ft^3.

Solving gives a loaded waterline $z_{wl} = 6.866$ in above the keel, displacing 12.034 ft^3 = 750.9 lb (matching the loaded weight, as required). Freeboard is the mold depth less that waterline:

$$\text{FB}_{\text{loaded}} = 15.07 - 6.866 = \mathbf{8.20} \text{ in} \tag{13}$$

against the 6.00 in minimum — **PASS** with 2.20 in of margin. Reserve buoyancy above the loaded waterline is 18.45 ft^3 (1151 lb).

Modelling note: above the design draft the offset table extrudes the waterline curve vertically (no flare). A flared hull would gain waterplane area as it sinks and float slightly higher, so the freeboard reported here is the conservative bound.